

PRATEEK KUMAR

Satna, Madhya Pradesh

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Experience

BITS Pilani (Goa)

August 2024 - January 2025

AIML Intern

Remote

- Accelerated CFD simulation result output from 40 minutes to 4 minutes (10x speedup) using deep learning
- Developed a deep learning ANN model that achieved 98% accuracy.
- Optimized the ANN algorithm to efficiently process 50,000+ data points and improved computational performance.
- Created 12+ data visualizations to track key performance metrics and support data-driven decision-making.
- Collaborated with cross-functional teams to develop real-time API analytics solutions.
- Leveraged Python ecosystem (TensorFlow, Pandas, NumPy) to build efficient data manipulation and analytical models.

Projects

Portfolio Website — GitHub Link — Live Demo

- Architected and deployed a full-stack portfolio using Next.js, TailwindCSS, and Framer Motion for an animated UI.
- Orchestrated a serverless backend with Prisma and PostgreSQL, integrating Resend API for real-time emails.
- Configured custom domain with Cloudflare and optimized SEO, performance, and accessibility for production readiness.

Content Catalyst — GitHub Link — Live Demo

- Developed an AI-powered platform for content creators to streamline media editing and enhancement.
- Integrated authentication and storage APIs for robust monetization pipeline, increasing API platform adoption by 40%.
- Set up webhook-based automation to streamline data flow and reduce latency by 65%.
- Enhanced content delivery with Cloudinary's video compression, achieving a 98% reduction in video size.
- Engineered scalable SaaS architecture using Docker containerization and robust API error handling systems.
- Technologies Used: *Next.js, TypeScript, daisyUI, Clerk, Cloudinary, Prisma, PostgreSQL, Docker Compose, Vercel*

End-to-End Twitter Depression Prediction System — GitHub Link

- Designed an API-driven depression prediction system with 90% accuracy, improving analytical insights.
- Implemented RESTful APIs for ML models, reducing response time by 30% and ensuring seamless integration.
- Automated data collection pipeline using Selenium, cutting manual extraction time by 60%.
- Built and fine-tuned a text classification model with spaCy and TensorFlow, increasing prediction accuracy by 15%.
- Technologies Used: *Python, Flask-RESTful, Selenium (Web Driver), React.js, spaCy, Scikit-learn, TensorFlow*

Education

Lakshmi Narain College of Technology, Bhopal

August 2021 - May 2025

Bachelor of Technology in Computer Science and Engineering (CGPA: 8.75/10)

Madhya Pradesh, India

- Relevant Coursework: DSA, DBMS, OOPs, Operating System, Machine Learning, Computer Vision.

St. Michael's Hr. Sec. School, Satna

May 2020

12th Grade (CBSE) (82.4%)

Madhya Pradesh, India

- Relevant Coursework: Physics, Chemistry, Mathematics, Computer Science (C++).

Technical Skills

- **Programming Languages:** C++, Python, JavaScript, TypeScript, SQL
- **Full-Stack Development:** Next.js, React.js, Express.js, Hono.js, Node.js, Django
- **Databases:** PostgreSQL, MongoDB, Prisma
- **DevOps & Cloud:** Git, GitHub, Docker, Docker Compose, Cloudflare Workers, AWS (S3, EC2), Nginx Reverse Proxy
- **AI/ML:** TensorFlow, Scikit-learn, Seaborn, spaCy, NLTK, Deep Learning, Computer Vision, LangChain

Achievements and Certifications

- Led teams in national-level hackathons such as Hackhive, Hackwave, and Flipkart Grid.
- Spearheaded a team of 4-5 members as GDSC AIML Lead, driving AI/ML initiatives and delivering 3+ events.
- Achieved an All India Rank of **6339** in GATE-DA 2024. [Link](#)
- Secured a position in the top **5%** merit list for NPTEL Discrete Mathematics Course by IIT Ropar.